

Water Electrolysis

Experiment Packet

Audience: instructors and teaching assistants

Purpose: Bench instructions for instructors and TAs

Session hook: Split water and see a 2:1 ratio

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Preparation
3. Materials checklist
4. Experiment procedure
5. Shared safety notes

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 4 — Water Electrolysis and Gas Collection

Hook: Tap water barely splits — add electrolyte, measure current, prove 2:1 stoichiometry, then pop an H₂ balloon.

Learning outcomes

- Explain why **tap water alone** electrolyzes poorly (few ions)
- Show that **Na₂SO₄** electrolyte increases current and reaction rate
- Measure **current** and estimate **H₂/O₂ production rate** with Faraday's law
- Relate applied **voltage** to the ~1.23 V minimum for water splitting
- Collect gases in a **U-tube** and confirm **~2:1 H₂:O₂** stoichiometry
- Identify H₂ with an instructor-led **balloon pop** demo
- **Start** electrolytic rust-removal cells for Session 5

Session plan (90 min)

Time	Activity	Reference
0–10 min	Hook: will graphite in tap water make gas? Why so slow?	lecture.md
10–20 min	Electrolyte, half-reactions, Faraday rate, 2:1 preview	lecture.md
20–30 min	Part A — Beaker + tap water only; record tiny current	experiment.md — Part A
30–45 min	Part B — Add Na ₂ SO ₄ ; measure I; calculate rate / discuss V	experiment.md — Part B
45–70 min	Part C — U-tube gas collection; volume ratio	experiment.md — Part C
70–80 min	Part D — Instructor H ₂ balloon fill + pop	experiment.md — Part D
80–90 min	Part E — Start derusting cells for Session 5	experiment.md — Part E

Key reactions

Overall: $2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$

Cathode (–): $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$

Anode (+): $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$ (acidic media)

or $4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$ (basic media)

Rate (Faraday): $\dot{n}(\text{H}_2) = I / (2F)$, $\dot{n}(\text{O}_2) = I / (4F)$

Status

- [] Beaker + graphite + multimeter stations ready
- [] ~0.1 M Na₂SO₄ mixed and labeled

- ☐ U-tube apparatus pilot-tested (2:1 visible)
- ☐ Balloon demo rehearsed or cancelled per venue rules
- ☐ **No saltwater** — Na_2SO_4 only for electrolysis
- ☐ Rusty nails + washing soda ready for Part E

Session 4 — Preparation

Before class (instructor)

- [] Pilot **Part A** → **B** in a beaker: record I (tap) vs I (Na_2SO_4) at fixed V
- [] Pilot Faraday arithmetic with those currents (students will copy your method)
- [] Pilot **U-tube** — confirm 2:1 ratio visible within session time
- [] Rehearse **H₂ balloon** demo (cathode-only feed, small balloon) or cancel per venue rules
- [] Check graphite rods, stoppers, spade connectors, multimeters
- [] Mix **~0.1 M Na₂SO₄** stock; label **"NO SALT"**
- [] Pre-weigh Na_2SO_4 portions (~3.5 g) for beaker additions if helpful
- [] Stage rusty nails, washing soda, sacrificial anodes, jars for **Part E**
- [] Confirm overnight power policy for derusting cells

Day-of setup

- [] Beaker stations: graphite, leads, series ammeter ready
- [] U-tube stations filled or ready to fill with Na_2SO_4
- [] Large sign: **"No salt in electrolysis — use sodium sulfate"**
- [] Balloon demo area marked; extinguisher present; long lighter staged
- [] Part E station ready for last 10 minutes

Common failure modes (from pilot)

Issue	Prevention
Part A "broken" complaint	Brief beforehand: tiny current is the point
Ammeter reads zero after Na_2SO_4	Wrong range; meter not in series; bad clip
Students grab salt	Pre-brief; hide NaCl; labeled Na_2SO_4 only
Balloon mixes gases	Dedicated cathode tube; anode vents free
Overpressure in U-tube	Side arms open

Open questions / notes

- Fixed voltage used for A/B comparison:
- Typical I_{tap} / $I_{\text{Na}_2\text{SO}_4}$:
- Balloon demo yes/no for this venue:

Session 4 — Materials

Per group — Parts A & B (beaker contrast)

- ☐ Beaker or clear cup (~200–400 mL)
- ☐ Two graphite rods
- ☐ DC power supply or battery pack + leads
- ☐ Multimeter for **current in series** (mA/A)
- ☐ Way to read **voltage** (second meter or supply display)
- ☐ Tap water
- ☐ **Na₂SO₄** to make ~0.1 M in the beaker (~3.5 g / 250 mL)
- ☐ Goggles

Per group / station — Part C (stoichiometry)

- ☐ Hoffman-style glass **U-tube** with side arms + acrylic stand
- ☐ Graphite electrodes with rubber/cork stoppers
- ☐ Red (+) / black (–) leads with spade connectors
- ☐ **~0.1 M Na₂SO₄** to fill the U-tube
- ☐ Ruler (gas-column / liquid-level comparison)

Instructor demo — Part D (H₂ balloon)

- ☐ Dedicated cathode feed (beaker/bottle + tubing) so only H₂ enters the balloon
- ☐ Small balloons
- ☐ Long lighter / candle on a stick
- ☐ Fire extinguisher accessible
- ☐ Clear demo area (or outdoor space per venue rules)

Part E — Start rust removal (end of class)

- ☐ Rusty iron nails or hardware (2+ per group)
- ☐ Washing soda: **1 tbsp per ~250 mL** warm water (preferred) **or** baking soda same measure (NOT NaCl)
- ☐ Sacrificial steel strips or graphite rods
- ☐ Small jars or cups
- ☐ Labels and camera for before photos

Optional

- ☐ pH / universal indicator
- ☐ Baking soda (emergency electrolysis backup only)
- ☐ Calculator / worksheet for Faraday rate

Explicitly excluded

- [] ~~Table salt (NaCl)~~ — **do not use** in electrolysis or derusting baths

Pilot notes

Parameter	Value from pilot
Applied V	__ V
I (tap water)	__ mA
I (Na ₂ SO ₄)	__ mA
Typical V _{H₂} /V _{O₂}	__
Balloon demo yes/no	

Session 4 — Experiment: Water Electrolysis

Instructors: electrolyte contrast, Faraday rate math, U-tube stoichiometry, H₂ balloon safety — instructor.md.

Session arc

1. **Part A** — Graphite in a beaker of **tap water only** (slow / almost no reaction)
2. **Part B** — Add **Na₂SO₄**; measure **current**; link voltage and rate to Faraday's law
3. **Part C** — Collect gases in the **U-tube**; measure volumes → **2:1 stoichiometry**
4. **Part D** — Separate demo: fill an **H₂ balloon** and pop it (instructor-led)
5. **Part E** — Start overnight **rust-removal** cells for Session 5

Electrolyte — sodium sulfate (Na₂SO₄)

Never use table salt (NaCl) — chloride can produce toxic chlorine gas at the anode.

Recipe — ~0.1 M Na₂SO₄

Batch	Na ₂ SO ₄ (anhydrous)	Water
250 mL	~3.5 g	Fill to 250 mL
500 mL	~7 g	Fill to 500 mL
1.0 L	~14 g	Fill to 1.0 L

Stir until dissolved. Label "**0.1 M Na₂SO₄ electrolyte — NO SALT**".

Water: start Parts A–B with plain **tap water** in the beaker; add solid Na₂SO₄ (or a concentrated aliquot) for Part B. Distilled water is unnecessary here.

Baking soda backup only if Na₂SO₄ is unavailable: 1–2 tsp NaHCO₃ per 500 mL.

Part A — Tap water only (slow reaction) (20–30 min)

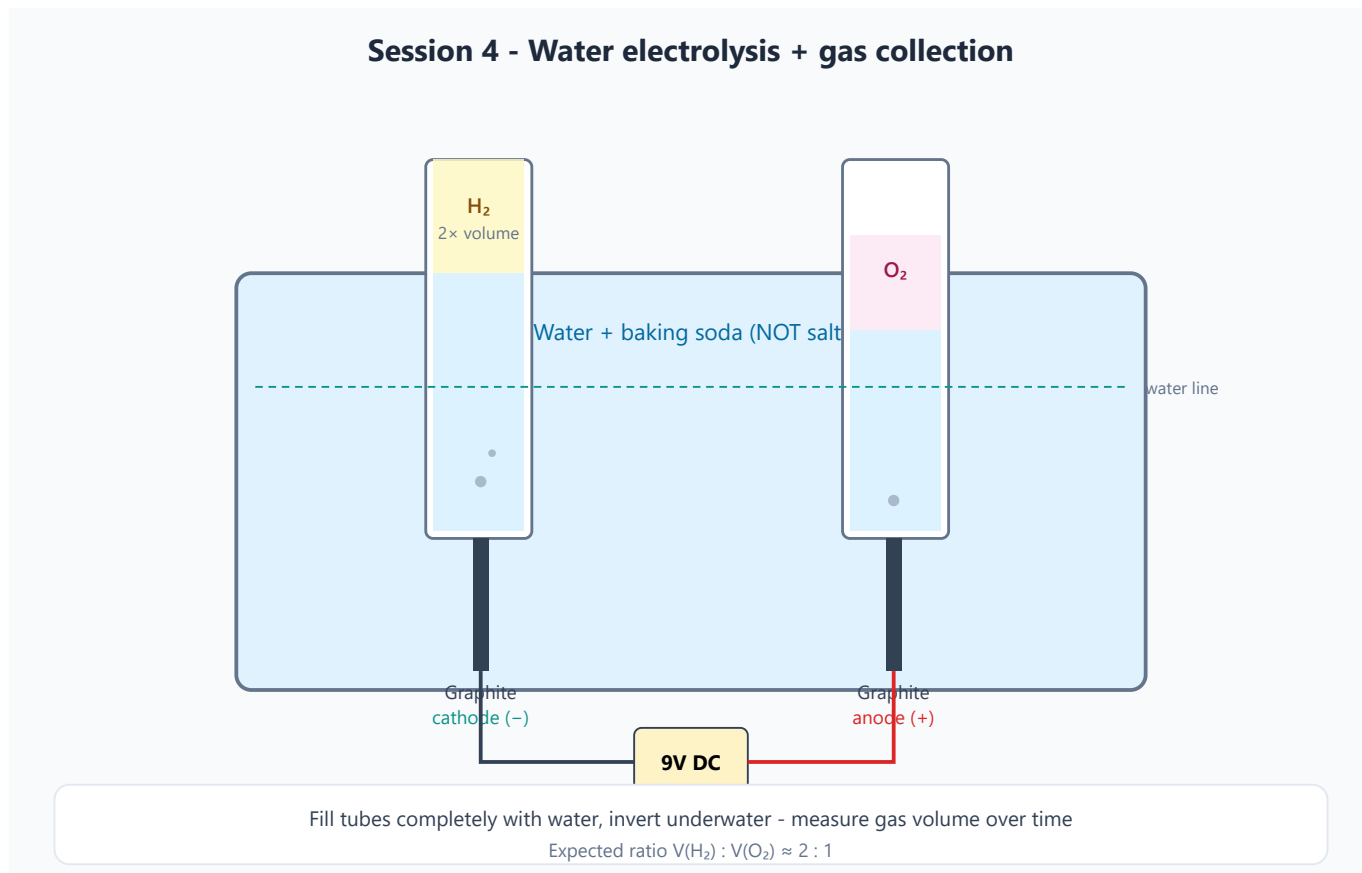


Figure 1 — Concept: H_2 at the cathode (-), O_2 at the anode (+). Without electrolyte, almost no current flows.

Materials (beaker cell)

- Beaker or clear cup (~200–400 mL)
- Two **graphite** rods (same rods you will use later in the U-tube if needed)
- DC supply or battery pack + leads (red +, black -)
- Multimeter (DC **mA** or A range) wired **in series**
- Multimeter or supply display for **voltage** across the cell
- Tap water only — **no salt, no Na_2SO_4 yet**
- Goggles

Procedure

1. Fill the beaker with **tap water** only (~200–300 mL).
2. Clamp or hold two graphite rods so they are immersed, **2–4 cm apart**, not touching.
3. Connect: black (-) to one rod (cathode), red (+) to the other (anode).
4. Put the ammeter **in series** with one lead. Measure voltage across the two electrodes (or read the supply setpoint).
5. Power on for **2–3 minutes**. Watch carefully for bubbles.

Expected observation

What you look for	Typical result (tap water)
Bubbles	None, or very few / slow
Current	Very small (often $\ll 1\text{--}10\text{ mA}$ depending on supply and meter)
Conclusion	Tap water has too few ions $\rightarrow$ poor conductor $\rightarrow$ slow electrolysis

Data — tap water

Quantity	Value
Applied voltage V (V)	
Current I (mA)	
Bubbles visible? (Y/N, describe)	
Time observed (min)	

Talking point: *Water itself is the reactant, but without mobile ions the circuit almost cannot run.*

Part B — Add electrolyte; measure current and rate (30–45 min)

Procedure

1. **Keep the same beaker and electrodes** (or reset with fresh tap water and the same spacing).
2. Add **Na_2SO_4** to make roughly **0.1 M** in the beaker:
3. For $\sim 250\text{ mL}$ water: add **$\sim 3.5\text{ g}$** anhydrous Na_2SO_4 ; stir until dissolved
4. Or pour in a pre-measured aliquot of stock 0.1 M Na_2SO_4 and top up
5. Reconnect the circuit with the ammeter in series. Use the **same applied voltage** as Part A if possible (fair comparison).
6. Record current once it stabilizes ($\sim 10\text{--}30\text{ s}$). Observe bubble rate at each electrode.
7. Optional: briefly raise/lower voltage and note how current (and bubbling) changes.

Expected observation

What you look for	Typical result (with Na_2SO_4)
Bubbles	Much faster at both electrodes
Current	Much larger than Part A
Conclusion	Electrolyte ions carry charge $\rightarrow$ higher I $\rightarrow$ faster water splitting

Data — with Na₂SO₄

Quantity	Value
Applied voltage V (V)	
Current I (mA) → convert to A: I = ___ A	
Qualitative bubble rate vs Part A	
Theoretical minimum for water splitting	~ 1.23 V (thermodynamic; real cells need more)

Calculations — voltage, current, and rate

1. Voltage needed (classroom framing)

- Ideal minimum to split water: **E° ≈ 1.23 V**
- Your supply must provide **at least** that, usually **more** (overpotential + resistance). Record the **applied V** you actually use.
- Optional: if the supply is adjustable, find the lowest V where bubbles become clearly visible.

2. Reaction rate from current (Faraday)

Charge of electrons: Faraday constant **F ≈ 96485 C/mol e⁻**

Hydrogen half-reaction uses **2 e⁻ per H₂**:

$$\text{rate of H}_2 \text{ production (mol/s)} = I / (2 F)$$

with **I in amperes** (A = C/s).

Oxygen uses **4 e⁻ per O₂**:

$$\text{rate of O}_2 \text{ production (mol/s)} = I / (4 F)$$

So the mole rate of H₂ is always **twice** the mole rate of O₂ for the same current — foreshadowing the volume ratio.

3. Optional volume rate at room conditions

At ~25 °C, 1 atm, molar volume ≈ **24.5 L/mol** (or use 22.4 L/mol at STP if your instructor prefers):

$$\begin{aligned}\text{rate of H}_2 \text{ (L/s)} &\approx [I / (2 F)] \times 24.5 \\ \text{rate of H}_2 \text{ (mL/min)} &\approx [I / (2 F)] \times 24.5 \times 1000 \times 60\end{aligned}$$

Worked example (fill in your I)

Step	Expression	Your value
I	___ A	
$\dot{n}(\text{H}_2) = I/(2F)$	___ mol/s	
$\dot{n}(\text{O}_2) = I/(4F)$	___ mol/s	

Step	Expression	Your value
Approx. H ₂ volume rate	___ mL/min	

Check: $\dot{n}(\text{H}_2) / \dot{n}(\text{O}_2)$ should equal **2** from the electron count alone.

Part C — Collect gas; measure stoichiometry (45–70 min)

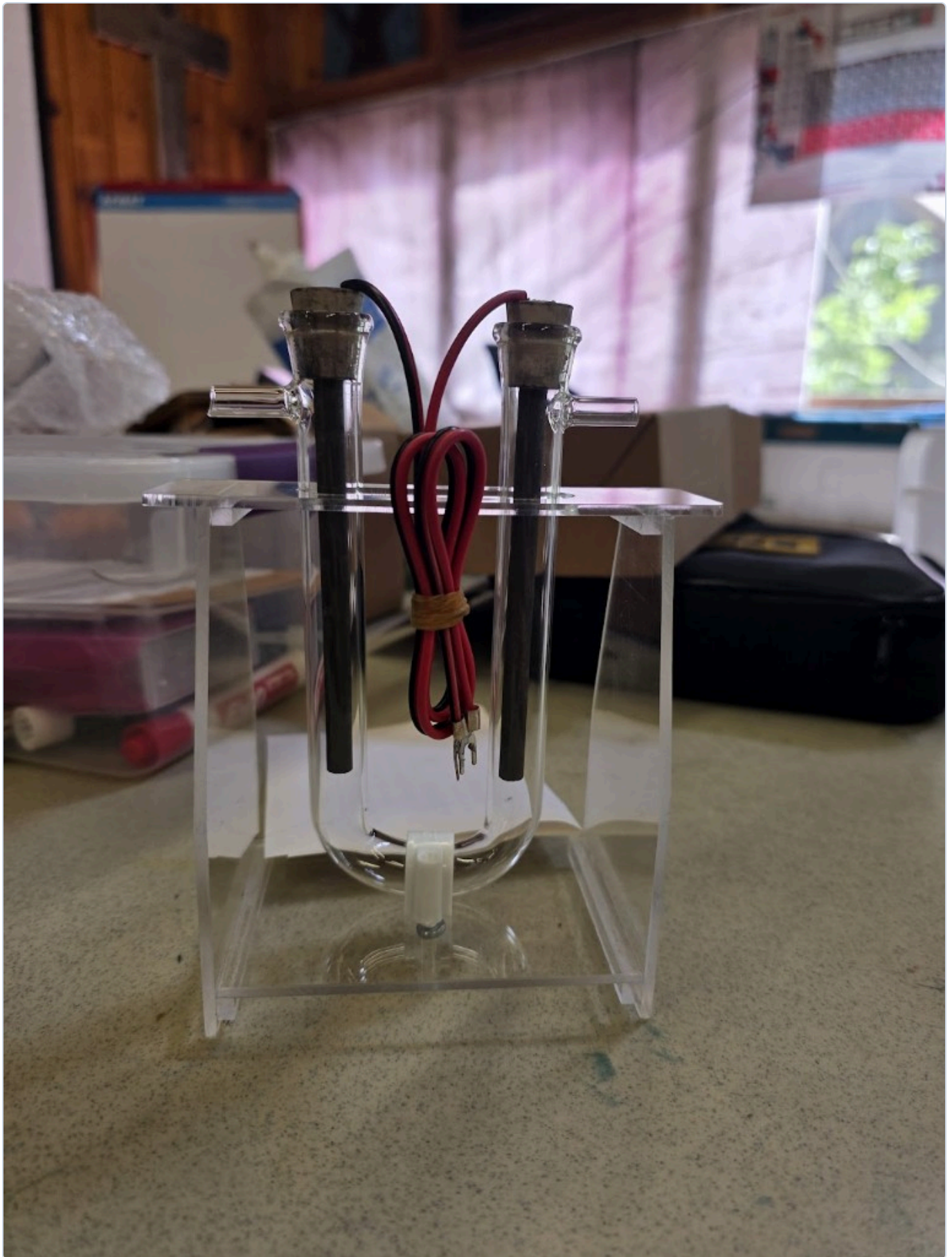


Figure 2 — U-tube apparatus for collecting H_2 and O_2 separately and comparing volumes.

Materials

- Hoffman-style **glass U-tube** with side arms + acrylic stand
- Graphite electrodes + rubber stoppers
- Red (+) / black (–) leads with spade connectors
- DC supply
- **0.1 M Na_2SO_4** (fill the U-tube)
- Ruler or limb graduations
- Goggles

Build / run

1. Fill the U-tube with **Na_2SO_4** electrolyte (liquid covers electrode tips; sits below side-arm ports).
2. Insert graphite electrodes through stoppers; seal tops; keep side arms **open or vented**.
3. Connect: **black (–) = H_2 limb, red (+) = O_2 limb**.
4. Power on. Mark starting liquid levels.
5. Every **2–3 min**, record gas column height or volume in each limb.

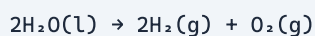
Gas volume log

Time (min)	V_ H_2 (mL or cm)	V_ O_2 (mL or cm)	Ratio V_ H_2 /V_ O_2
0	0	0	—
3			
6			
9			
12			
15			

Expected result

V_ H_2 / V_ O_2 $\approx$ **2** (ideal). Classroom range **1.6–2.2** is a success.

Connect to Part B: the same **2:1** appears in Faraday rates **and** in collected volumes — that is the stoichiometry of



Pre-run checklist

- [] Electrolyte is **Na_2SO_4** — **not salt**
 - [] Polarity marked; side arms vented
 - [] Goggles on
-

Part D — H₂ balloon fill and pop (70–80 min)

Instructor-led demo (students observe; do not hand students lit flames). Use a **separate** cathode setup from the U-tube if needed so stoichiometry data are already recorded.

Idea

Collect hydrogen from the **cathode (-)** into a small balloon, then ignite for a loud **pop** — qualitative proof that the fast gas is combustible H₂.

Materials (demo station)

- Beaker or bottle with **Na₂SO₄** + two graphite electrodes (or cathode tubing from a dedicated H₂ generator)
- Small balloon (party balloon)
- Tubing / stopper so H₂ from the cathode feeds the balloon (O₂ must **not** mix into the balloon)
- Long lighter / candle on a stick
- Fire extinguisher nearby; clear area; eye protection for everyone nearby

Procedure (instructor)

1. Set up electrolysis so **only cathode gas** enters the balloon (anode vents separately to air).
2. Run until the balloon is **modestly** inflated (small volume — do not make a large balloon).
3. Tie off; move **away** from the electrolysis bath and from faces.
4. Ignite with a long lighter at arm's length → sharp pop.
5. Never ignite a balloon that may contain mixed H₂/O₂; never use a sealed rigid container.

Safety (non-negotiable)

- [] Instructor only handles flame
 - [] Small balloon only; outdoor or well-cleared indoor space per venue rules
 - [] No salt electrolyte
 - [] Skip entirely if open flame is forbidden
-

Part E — Start electrolytic rust removal (80–90 min)

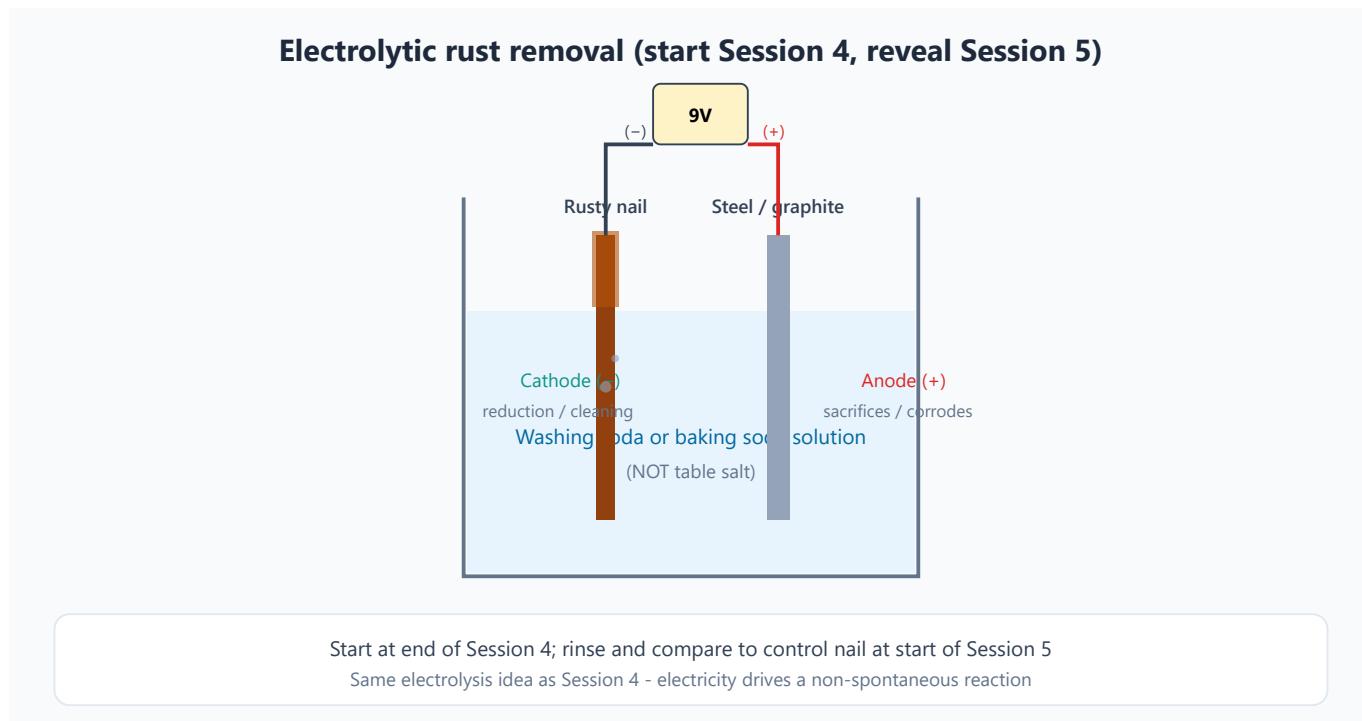


Figure 3 — Start cells now; reveal cleaned nails at opening of Session 5.

Derusting electrolyte

Ingredient	Amount
Washing soda (Na_2CO_3) preferred	1 tbsp (~15 g) per ~250 mL warm water
or baking soda	1 tbsp per ~250 mL warm water

Do NOT use NaCl. Separate bath from the Na_2SO_4 electrolysis.

Setup steps

1. Dissolve washing soda (or baking soda); pour into a small jar.
2. Rusty iron → battery **cathode (-)**.
3. Sacrificial steel or graphite → battery **anode (+)**.
4. Immerse without touching; confirm gentle bubbling.
5. Label; photograph **before**; set a control nail in plain water (no battery).
6. Leave powered overnight per venue rules.

End-of-class checklist

- [] Derusting cells bubbling / current confirmed
- [] Controls labeled; before photos taken
- [] Overnight power policy confirmed

Troubleshooting

Problem	Cause	Fix
No bubbles in Part A	Expected if current tiny	Proceed to add Na_2SO_4
Still no bubbles in Part B	Bad contacts / wrong polarity	Check clips; raise V slightly; remake ~0.1 M Na_2SO_4
Current barely rises after salt... wait, Na_2SO_4	Not dissolved / too dilute	Stir; add more sulfate (not NaCl)
Chlorine smell	Salt used	Stop; dump; remake Na_2SO_4
Ratio far from 2:1	Leaks; late start; unequal arms	Reseat stoppers; restart after steady bubbling
Balloon won't inflate	Leak; wrong electrode to balloon	Confirm cathode feed only
Graphite crumbling	High current / long run	Lower voltage; replace rods

Safety checklist

- ☐ No NaCl in any electrolysis bath
 - ☐ Ammeter in **series** (never short the supply)
 - ☐ Flames only for instructor balloon / tiny pop demo
 - ☐ U-tube side arms vented (no overpressure)
 - ☐ Goggles on for wet work
-

Experiment status

- ☐ Part A/B beaker contrast pilot-tested (I before vs after Na_2SO_4)
- ☐ Faraday rate worksheet numbers checked with pilot current
- ☐ U-tube 2:1 visible in pilot
- ☐ Balloon demo rehearsed safely (or cancelled per venue)
- ☐ Rust-removal jars staged for Part E

Syllabus Safety Notes

Apply these across all sessions. Session-specific hazards are listed in each session folder.

General rules

1. **No eating experimental materials.** Fruits, potatoes, and liquids used in experiments are no longer food.
2. **Avoid short circuits.** Batteries can heat up if directly shorted.
3. **Clean surfaces matter.** Electroplating requires clean metal surfaces.
4. **Metal salt waste should be collected.** Copper and silver solutions should not be casually poured down the drain.
5. **Water choice:** tap water is fine for most baths; use **distilled/deionized water for 0.01 M AgNO₃** (Session 3) so chloride does not precipitate silver.

Session-specific hazards

Session	Key hazard	Mitigation
1	Sharp nails/wires	Supervise handling; wash hands after metals
2	Copper solutions	Goggles; no ingestion; collect waste
3	Silver nitrate	Gloves + goggles; stains skin/clothing; small volumes
4	Gas evolution + H ₂ balloon	No saltwater; ammeter in series; instructor-only balloon/flame demo
5	Corrosion demos + electrolytic derusting	Use washing soda or baking soda only — NOT NaCl; disconnect power when not supervised

Hydrogen balloon / pop demo (Session 4, instructor-only)

- Feed **cathode (H₂) gas only** into a **small** balloon — do not mix with O₂
- Inflate modestly; move away from the electrolysis bath and faces
- Long lighter at arm's length; clear area; extinguisher accessible
- **Never** ignite a sealed rigid container or a mixed H₂/O₂ balloon
- Skip entirely if the venue forbids open flame